INCH-POUND

MIL-C-70562B (AR)
8 December 1992
SUPERSEDING
(See 6.6)

MILITARY SPECIFICATION

COMPUTER, RANGE CP-888A/VPS-2A: 9361153

This specification is approved for use by the U. S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, examinations and tests for a Computer, Range CP-888A/VPS-2A, hereinafter referred to as the computer. The computer is a part of Radar Set AN/VPS-2A. The radar set range computer generates range and test data based on the signals received from the receiver. The range computer then transmits this information to the fire control processor where it is used in solving the fire control problem. The radar set range computer contains a built-in-test circuit assembly for signal monitoring and fault isolation (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (See 6.2).

SPECIFICATIONS

MILITARY

MIL-F-13926 - Fire Control Materiel, Manufacture and Inspection, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
FSC 1285

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MIL-I-45607 - Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-P-50317 - Protector, Over Voltage: 10548031 (+5 V Filter)
MIL-R-50318 - Regulator, Voltage: 10548034 and 11727766
MIL-R-50319 - Regulator, Voltage: 10548043 and 11727763
MIL-G-50322 - Generator, Digital: 10548613
MIL-C-50344 - Counter, Electronic: 10548009
MIL-C-50652/1 - Modulator Trigger Generator: 9355881
MIL-C-50652/2 - Computer Timing Generator: 10548007
MIL-C-50652/3 - RCFD Decoder: 10548119
MIL-C-50652/4 - Range Counter: 10548046
MIL-C-50652/5 - RCFD Generator: 10548016
MIL-C-50652/6 - Range Register A: 10548122
MIL-C-50652/7 - Range Register B: 10548021
MIL-C-50652/8 - Search Track and Gain Control: 10548148
MIL-C-50652/9 - Range Rate Counter A: 10548125
MIL-C-50652/10 - Range Rate Counter B: 10548012
MIL-C-50652/11 - Rate Scaler: 10548049
MIL-C-50652/12 - Rate Integrator: 10548086
MIL-C-50652/13 - Phase Detector: 10548133
MIL-C-50652/16 - Control Amplifier: 10548450
MIL-C-50652/17 - Fine/Fine Delay: 10548638
MIL-C-50652/18 - Universal Board: 10549234
MIL-R-70560 - PIVADS Radar Set AN/VPS-2A: 9360797
MIL-C-70571 - Circuit Board, Built-In-Test #1, Unit 4 (PIVADS): 9360949
MIL-C-70572 - Circuit Board, Built-In-Test #2, Unit 4 (PIVADS): 9360952

STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

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DRAWINGS (See 6.4)

U. S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9361153 - Computer, Range CP-888A/VPS-2A
9361217 - Remote BIT Tester Assembly
9361225 - W303 Cable, Unit 4.
12950877 - Requirements for Soldering Electrical and Electronic Assemblies

SPECIAL PACKAGING INSTRUCTIONS

PS9361331 - Packaging of Computer, Range CP-888A/VPS-2A W/CASE

(Copies of drawings, publications and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of conflict between the text of this document and the references cited herein the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 9361153, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the computer shall meet the requirements of MIL-F-13926 as follows:

a. Dimensions and tolerances.
b. Inorganic prctective surface finishings.
c. Part identification and marking.
d. Workmanship.

3.2.1 Subassemblies. The subassemblies of the computer shall have been inspected and accepted in accordance with their respective drawings and specifications prior to their installation in the computer. The following subassemblies are covered by the indicated specifications:

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UNIT NO.	NAME	SPECIFICATION
A1	Generator Digital: 10548613	MIL-G-50322
A2	Counter Electronic: 10548009	MIL-C-50344
A3	Modulator Trigger Generator: 9355881	MIL-C-50652/1
A4	Computer Timing Generator: 10548007	MIL-C-50652/2
A5	RCFD Decoder: 10548119	MIL-C-50652/3
A6	Range Counter: 10548046	MIL-C-50652/4
A7	RCFD Generator: 10548016	MIL-C-50652/5
A8	Range Register A: 10548122	MIL-C-50652/6
A9	Range Register B: 10548021	MIL-C-50652/7
A10	Search, Track and Gain Control: 10548148	MIL-C-50652/8
A11	Range Rate Counter A: 10548125	MIL-C-50652/9
A12	Range Rate Counter B: 10548012	MIL-C-50652/10
A13	Rate Scaler: 10548049	MIL-C-50652/11
A14	Rate Integrator: 10548086	MIL-C-50652/12
A15	Phase Detector: 10548133	MIL-C-50652/13
A16	BIT Circuit Board 2: 9360952	MIL-C-70572
A17	BIT Circuit Board 1: 9360949	MIL-C-70571
A18	Protector, Overvoltage: 10548031 (+5 V Filter)	MIL-P-50317
A19	Regulator, Voltage: 10548034 and 11727766	MIL-R-50318
A20	Regulator, Voltage: 10548043 and 11727763	MIL-R-50319
A21	Control Amplifier: 10548450	MIL-C-50652/16
A22	Fine/Fine Delay: 10548638	MIL-C-50652/17
A23	Universal Board: 10549234	MIL-C-50652/18

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3.3 Environmental service conditions.

3.3.1 Vibration. The computer shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to the equivalent of 6000 miles of track-laying vehicle operational vibration as defined by MIL-STD-810 narrow band random-on-random data for equipment installed on the deck of the M113 armored personnel carrier.

3.3.2 Shock. The computer shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to a total of 18 impulse shocks consisting of three shock impulses of 15 g's ± 10%, 11 milliseconds ± 10%, half-sine wave applied in each direction of 3 mutually perpendicular axes.

3.3.3 Altitude operating. The computer shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches of Hg).

3.3.4 Temperature.

3.3.4.1 Operating. The computer shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between -40 deg.F and +155 deg.F.

3.3.4.2 Non-operating (storage). The computer shall not suffer permanent damage when subjected to the constant temperature extremes of -60 deg.F for a period of 24 hours and +160 deg.F for a period of 4 hours (after temperature stabilization), and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (non-operating). The computer shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram of the computer.

3.5 Performance. For the requirements of 3.5.1 through 3.5.24, the following conditions shall apply:

a. Logic 0: 0.7 V Max
b. Logic 1: 2.4 V Min
c. Positive pulse: A change in dc level from 0.0 ± 0.5 V to +3.5 ± 1.0 V and return to 0.0 ± 0.5 V.
d. Negative pulse: A change in dc level from +3.5 ± 1.0 V to 0.0 ± 0.5 V and return to +3.5 ± 1.0 V.
e. Switch 4S2 (Mode): In test position.

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f. Switch 4S3 (Accel): In negative position.
g. Switch 4S4 (Return): In no target position.
h. Switch 4S5 (Range): In run position.
i. Switch 4S6 (Rate): In normal position.
j. Delays in signals when specified shall be from the leading edges of each signal.
k. All pulse widths are specified at the 50 percent points.

3.5.1 Voltage inputs and distribution. With the input voltages applied as specified in Table I (Input voltages), the internal voltages for distribution to the assemblies in the computers shall be as specified in Table II (Distributed voltages).

TABLE I. Input voltages.

INPUT VOLTAGES (Vdc)	CHARACTERISTICS (Vdc)	TERMINALS
-33	-33.0 ± 0.25	J1-a(-), J1-b(RET)
+33	+33.0 ± 0.25	J1-Y(+), J1-Z(RET)
+6.5	+6.5 ± 0.5	J1-e(+), J1-g(RET)
		J1-d(+), J1-f(RET)
+24	+24.0 ± 0.5	J1-T(+), J1-U(RET)

TABLE II. Distributed voltages.

VOLTAGE (Vdc)	CHARACTERISTICS (Vdc)	TERMINAL TEST POINT
-25	-25.0 ± 0.5	E1(-), E2(RET)
+25	+25.0 ± 0.5	E3(+), E4(RET)
+5	+5.0 ± 0.5	E9(+), E10(RET)
		W1(+), W2(RET)
-15	-15.0 ± 0.5	E5(-), E6(RET)
+15	+15.0 ± 0.5	E8(+), E7(RET)

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3.5.2 Midrange calibration lamp, 4DS1. With voltages applied as specified in Table I and no input signals applied, the midrange calibration lamp 4DS1 shall be dimly lit.

3.5.3 Midrange calibration switch, 4S1. With +24 Vdc applied to 4J1-T and with "PRESS TO TEST" switch 4S1 depressed, there shall be +24 Vdc present at 4J1-R and 4J3-C.

3.5.4 7.5 MHz clock output. With 60.0 ± 5.0 megahertz (MHz) having an amplitude between the limits of 2.5 to 4 V peak-to-peak applied to J4-A, a 7.5 ± 0.6 MHz clock signal shall be present at TP38 on the computer. The clock signal shall have a period of 133 ± 10 nanoseconds (ns) with an amplitude of 4.0 V peak-to-peak or greater.

3.5.5 $\overline{\text{PRF RESET}}$ signal. The $\overline{\text{PRF RESET}}$ signal at TP20 shall be a negative pulse, 67 ± 25 ns wide, occurring every 50 ± 5 microseconds (microsec).

3.5.6 $\overline{\text{PRF}}$ signal. The $\overline{\text{PRF}}$ signal at TP19 shall be a negative pulse having the same characteristics as specified in 3.5.5.

3.5.7 $\overline{\text{PRF CAL}}$ signal. With 4S1 midrange calibration switch depressed, the $\overline{\text{PRF CAL}}$ signal at TP19 shall be two negative pulses. The pulses shall be separated by 67 ± 25 ns. The first pulse shall be delayed 16 ± 2 microsec from the signal specified in 3.5.5.

3.5.8 $\overline{\text{PRF}}$ signal. The $\overline{\text{PRF}}$ signal at TP31 shall be a negative pulse, 0.6 ± 0.1 microsec wide, occurring every 50 ± 5 microsec. With switch 4S1 depressed, the negative pulse at TP31 shall be 0.8 ± 0.1 microsec wide, occurring every 50 ± 5 microsec, and be delayed 16 ± 2 microsec from the signal specified in 3.5.5.

3.5.9 $\overline{\text{T1}}$ signal. The $\overline{\text{T1}}$ signal at TP21 shall be a positive pulse, 0.75 ± 0.10 microsec wide occurring every 50 ± 5 microsec, and be delayed 34 ± 4 microsec from the signal specified in 3.5.5.

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3.5.10 MOD To TRIGGER signal. The MOD To TRIGGER signal at TP45 and 4J4-E shall be a positive pulse of not less than 17 V peak and having a maximum rise time, of 100 ns into a 150 ohm load. Any pulse ringing shall not recross a 15.5 V threshold. The signal at TP45 shall be delayed 150 ± 30 ns from the signal specified in 3.5.8, measured from rising edge to rising edge. The signal shall show no changes when switch 4S1 is depressed.

3.5.11 T2 signal. The T2 signal at TP34 shall be as specified in 3.5.9 except the delay shall be 1.0 ± 0.1 microsec from the T1 signal specified in 3.5.9.

3.5.12 T3 signal. The T3 signal at TP23 shall be as specified in 3.5.9 except the delay shall be 1.0 ± 0.1 microsec from the T2 signal specified in 3.5.11.

3.5.13 $\overline{\text{T11}}$ signal. The $\overline{\text{T11}}$ signal at TP32 shall be a negative pulse, 0.75 ± 0.10 microsec wide occurring every 50 ± 5 microsec, and delayed 43 ± 3 microsec from the $\overline{\text{PRF RESET}}$ signal specified in 3.5.5.

3.5.14 $\overline{\text{T1/250}}$ signal. The $\overline{\text{T1/250}}$ signal at TP9 shall be a complementary pulse (negative pulse) to the T1 signal as specified in 3.5.9 except the $\overline{\text{T1/250}}$ pulse shall occur every 4.0 ± 0.5 microsec.

3.5.15 $\overline{\text{T3/4KHz}}$ signal. The $\overline{\text{T3/4KHz}}$ signal at TP10 shall be a coincident pulse to the T3 signal as specified in 3.5.12 except the $\overline{\text{T3/4KHz}}$ pulse shall occur every 250 ± 25 microsec.

3.5.16 $\overline{\text{200M}}$ signal. With switch 4S4 in the target position, the $\overline{\text{200M}}$ signal at J8-55 shall be a logic 1 level. With switch 4S4 in the target position and 24 Vdc applied between 4J1-G(-) and 4J1-H(RET), the $\overline{\text{200M}}$ signal at J8-55 shall be a logic 0 level.

3.5.17 T3 --> T10 serial and $\overline{\text{R}}$ serial signals. With switches 4S2 through 4S6 sequentially placed as specified in Table III (Placement of switches), the T3 --> T10 serial and $\overline{\text{R}}$ serial signals at TP17 and TP18, respectively, shall be as specified in Table III. Signal waveform characteristics are shown in Figure 2. When going from step 1 to step 2 of Table III, the signal at TP18 as shown in Figure 2 (B) changes to that shown in Figure 2 (C) and all eight time slots of the signal shall be counting, i.e., no pulses shall remain always present or always missing.

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TABLE III. Placement of switches.

STEP	4S2 MODE	4S3 ACCEL	4S4 RETURN	4S5 RANGE	4S6 RATE	SIGNAL CHARACTERISTICS TP17	TP18
1	Test	Pos.	Tgt.	Run	Normal	Figure 2a	Figure 2b
2	Test	Neg.	Tgt.	Run	Normal	Figure 2a	Figure 2c
3	Test	Pos.	Tgt.	Run	Reset	Figure 2a	Logic 0
4	Test	Neg.	Tgt.	Run	Reset	Figure 2a	Logic 0
5	Test	Pos.	No tgt.	Run	Normal	Figure 2a	Logic 0
6	Test	Neg.	No tgt.	Run	Normal	Figure 2a	Logic 0

3.5.18 TRACK signal.

3.5.18.1 No target condition. With switches 4S2 through 4S6 placed in positions as specified in 3.5, the TRACK signal at TP15 and TP24 shall be a logic 0 level.

3.5.18.2 Target condition. When switch 4S4 is changed from the no target position to the target position, the TRACK signal at TP15 and TP24 shall change from a logic 0 level to a logic 1 level.

3.5.18.3 Target fade (short term). When switch 4S4 is changed from the target position to the no target position, the TRACK signal at TP15 and TP24 shall change from a logic 1 level to a logic 0 level after 1.00 $\pm$ 0.20 second(s).

3.5.18.4 Target fade (long term). When switch 4S6 is placed in the reset position and simultaneously switch 4S4 is changed from the target position to the no target position, the TRACK signal at TP15 and TP24 shall change from a logic 1 level to a logic 0 level after 3.00 $\pm$ 0.25 s.

3.5.18.5 Target lock-on signal. When switch 4S4 is changed from the no target position to the target position, the target lock-on signal between 4J5-C (signal) and 4J5-D (signal return) shall change from a logic 1 level to a logic 0 level.

3.5.19 MIDRANGE lamp 4DS1. With 4S4 in the "No Target" position, the midrange lamp shall be flashing brightly at 1 second intervals. The lamp flashes when not in track.

3.5.20 E/L COARSE signal. The E/L COARSE signal at TP33 shall be a single positive-going step function, whose leading edge moves from less than 2.0 microsec to 33 $\pm$ 3 microsec with respect to the PRF signal at TP31, at 1 second intervals.

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3.5.21 PRE RANGE TRIGGER. The PRE RANGE TRIGGER signal at TP36 and 4J4-B shall be a single positive-going step function whose leading edge moves in synchronization with, and less than 2 microsec behind, the E/L COARSE signal at TP33.

3.5.22 E/L TRIGGER.

3.5.22.1 E/L TRIGGER characteristics. The E/L TRIGGER signals shall be present at TP43, TP44 and 4J4-D, and have the characteristics as specified in Table IV (E/L TRIGGER characteristics).

TABLE IV. E/L TRIGGER characteristics.

TERMINAL	
TP43	Positive pulse 0.4 to 1.5 microsec wide, and moves from a delay of less than 2 microsec to a delay of 33 ± 3 microsec with respect to the signal at TP31.
TP44	Positive pulse 0.2 to 0.6 microsec wide, and moves with the same delays as specified for TP43.
4J4-D	Positive pulse with the same characteristics as the signal at TP44 except that it is delayed by 1.0 ± 0.3 microsec from the signal at TP44.

3.5.22.2 E/L TRIGGER movements. When switches 4S3 through 4S6 are sequentially positioned as specified in Table V (E/L TRIGGER movement), the E/L TRIGGER signals at TP43 and TP44 shall have the characteristics as specified in Table IV except the direction and times of movement shall be as specified in Table V.

TABLE V. E/L TRIGGER movement.

	SWITCH POSITIONS				SIGNAL CHARACTERISTICS
STEP	4S5	4S3	4S4	4S6	TP43, TP44
1	Run	Neg.	No tgt.	Normal	Left to right motion at 1 s rate.
2	Freeze	Neg.	No tgt.	Normal	No motion.
3	Freeze	Pos.	Tgt.	Normal	No motion.

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TABLE V. E/L TRIGGER movement (continued).

	SWITCH POSITIONS				SIGNAL CHARACTERISTICS
STEP	4S5	4S3	4S4	4S6	TP43, TP44
4	Run	Pos.	Tgt.	Normal	Left to right motion at a 17 ± 1 s rate.
5	Run	Neg.	Tgt.	Normal	Right to left motion at a 17 ± 1 s rate.
6	Run	Toggle switch between pos. & neg. several times	Tgt.		Pulse motion shall be faster (almost twice as fast) for the first few seconds after each toggle of 4S4, then resume the 17 s rate.
7	Run	Same as step 6	Tgt.	Reset momen-tarily for each toggle	Pulse motion shall cease until 4S6 is placed back to normal, then pulse motion shall be double for first few seconds, then resume the 17 s rate.

3.5.22.3 E/L TRIGGER leading edges. The E/L TRIGGER signals at TP43, TP44 and 4J4-D shall have leading edges with the characteristics as specified in Table VI (E/L TRIGGER leading edges); switches 4S2 through 4S6 shall be as specified in 3.5.

TABLE VI. E/L TRIGGER leading edges.

TEST POINT OR TERMINAL	TEST POINT FOR SYNCHRONIZATION	CHARACTERISTIC
TP43	TP33	Sixteen leading edges spaced at 8.2 ± 1.4 nsec (± 2.4 ns between edges 8 and 9). The total duration from first to last leading edge shall be 123 ± 10ns.

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TABLE VI. E/L TRIGGER leading edges. (continued)

TEST POINT OR TERMINAL	TEST POINT FOR SYNCHRONIZATION	CHARACTERISTIC
TP44	TP43	Five leading edges evenly spaced 1.7 ± 0.2 ns. The total duration from first to last leading edge shall be 6.5 ± 0.5 ns.
4J4-D	TP43	Five leading edges with the same characteristics as for TP44 except with the delay as specified for 4J4-D in Table IV.

3.5.23 $\dot{R}>$ ± 10 M/S signal. When the steps of Table VII ($\dot{R}>$ ± 10 M/S signal) are sequentially performed, then $\dot{R}>$ ± 10 M/S signal at TP42 shall be as specified in Table VII under the following conditions:

a. 4S2 placed in Test.
b. 4S3 placed as specified in Table VII.
c. 4S4 placed in Target.
d. 4S5 placed in Run.
e. 4S6 as specified in Table VII.

TABLE VII. $\dot{R}>$ ± 10 M/S signal.

STEP	4S3	4S6	TP42 SIGNAL CHARACTERISTICS
1	Neg.	Run	Logic 0 level.
2	Neg.	Reset	Logic 1 level.
3	NOTE	Run	Momentary transition from logic 0 to logic 1 and back to logic 0 level again each time 4S3 is toggled.

NOTE: Toggle 4S3 between the negative and positive positions several times.

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3.5.24 Doppler signals response. The computer shall respond to the application of sum and difference signals sequentially applied as specified in Table VIII (Doppler signal response) with the following conditions:

a. Sum and difference input signal characteristics as specified in Table IX (Doppler input and output signals).

b. The E/L trigger at TP44 shall move as specified in Table VIII between the delay limits of less than 2 microsec to 33 ± 3 microsec with respect to the signal at TP45.

c. The ECM relay (4K3) shall provide contact closures between terminal 4J1-B and that specified in Table VIII.

d. The TGT lock-on relay (4K2) shall provide contact closures between terminal 4J1-D and that specified in Table VIII.

e. The doppler audio output test signal shall be as specified in Table IX for the conditions of step 4 of Table VIII.

TABLE VIII. Doppler signal response.

STEP	TP44 SIGNAL	ECM RELAY CLOSURE	LOCK-ON RELAY	SUM SIGNAL P-P	DIFFERENCE SIGNAL P-P	PHASE LAG OF DIFFERENCE
1	Left to right, 1 s	4J1-A	4J1-C	No signal	No signal	No signal
2	Left to right, 1 s	4J1-A	4J1-C	No signal	1.5 V	170°
3	Right to left, 17 s	4J1-A	Note 1	0.2 V	0.75 V	170°
4	Left to right, 17 s	4J1-A	4J1-w	0.2 V	0.75 V	10°
5	Stationary (Note 2)	4J1-A	4J1-w	0.2 V	0.75 V	10°

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TABLE VIII. Doppler signal response. (continued)

STEP	TP44 SIGNAL	ECM RELAY CLOSURE	LOCK-ON RELAY	SUM SIGNAL P-P	DIFFERENCE SIGNAL P-P	PHASE LAG OF DIFFERENCE
6	Left to right, 17 s	4J1-A	4J1-w	0.2 V	0.75 V	10°
7	Stationary (Note 2)	Note 3	4J1-w	1.1 V	0.75 V	10°

NOTES: 1. Adjustable to point where contact closure for terminal 4J1-w shall occur when the 0.2 V sum input level is reached.
2. With the RATE switch (4S6) in the reset position, the E/L trigger signal shall stop moving and remain stationary.
3. Adjustable to point where contact closure for terminal 4J1-V shall occur when 1.1 V sum input level is reached.

TABLE IX. Doppler input and output signals.

REQUIREMENT	CHARACTERISTIC	TERMINAL
Sum signal	Reference phase: 1.0 ± 0.1 KHz sine wave, amplitude in peak-to-peak volts as specified in Table VIII and as measured between TP47 (signal) and TP46 (signal return) on the computer.	4J4-F
Difference signal	Variable phase: 1.0 ± 0.1 KHz sine wave having a lagging phase angle from the reference phase as specified in Table VIII, amplitude in peak-to-peak volts as specified in Table VIII and as measured between TP49 (signal) and TP48 (signal return) on the computer.	4J4-G
Audio signal	Sine wave having same characteristics as the sum signal except the amplitude shall be adjustable to 1.0 ± 0.1 V peak-to-peak under conditions in step 6 of Table VIII.	4J1-E (signal) 4J1-F (signal return)

3.5.25 Subclutter visibility and range accuracy tests. The computer after having met the requirements of 3.5.1 through 3.5.24, shall meet the subclutter visibility (SCV) and range accuracy tests as specified in MIL-R-70560 when substituted into a Radar Set AN/VPS-2A. Radar Set AN/VPS-2A and facilities shall be as specified in MIL-R-70560.

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3.5.26 Built-In-Test (BIT) assembly. The computer shall contain a remote BIT assembly for transmitting range and test data to the fire control processor, signal monitoring and fault isolation. During RADAR mode, the BIT assembly shall convert parallel digital range data to several format and continuously transmit the data to the FCP in a 16-bit Range/BIT word (RBW). During TEST mode, the BIT assembly shall transmit the RBW with any one of 4 types of data, based on the format requested by the FCP. The 4 types of data shall be as follows:

a. Range and mode status (coast, search, track or ECM) data.
b. Range rate and gain data.
c. Digital test point data, MRC status and Unit 4 test switches status.
d. Sampled input and output voltage values from 4A19 and 4A20 regulators and the 4A18 Filter Card.
The BIT assembly shall be composed of two circuit boards, Bit Board 1 and Bit Board 2, interconnected by a ribbon cable.

3.5.26.1 BIT Board 1 (4A17). BIT Board 1 is 5-1/2 x 5 inch, 4-layer printed circuit board through which range and test data is transmitted to the fire control processor. BIT Board 1 performance shall be in accordance with MIL-C-70571.

3.5.26.2 BIT Board 2 (4A16). BIT Board 2 is 4-1/2 x 5 inch, 4-layer printed circuit board through which test data and range data is transmitted to the fire control processor. BIT Board 2 performance shall be in accordance with MIL-C-70572.

3.5.27 Voltage and signal outputs. With the input voltages applied as specified in Table I, and the remote bit test assembly PN 9361217 connected to J2 (Ref Fig 4), the conditions of Table XIV shall be met.

3.5.28 Switch Controls. With the input voltages applied as specified in Table I, -0.5 $\pm$ 0.1 VDC applied to terminals 4J4-F and 4J4-G and remote bit test assembly PN9361217 connected to P2 (Ref Fig 3), the conditions of Table XV shall be met.

3.6 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.7 Soldering. All soldering shall be in accordance with the requirements of Note 2 of Drawing 12950877.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinaitons and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Special sampling inspection (see 4.5).

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature	+50° to +90°F
b.	Relative humidity	90% RH Max.
c.	Pressure	28 to 32 inches of Hg

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4.3 First article inspection.

4.3.1 First article submission. The contractor shall submit a first article sample as designated by the contracting officer consisting of three (3) computers for inspection in accordance with 4.3.2.

4.3.2 Inspections to be performed. First article assemblies, components and test specimens may be subjected by the Government to any or all of the examinations and tests specified in Table X (Functional test) and Table XI (Special sampling) herein and to any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly, component or test specimen to comply with any of the stated requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests.

4.4.2.1 Components, assemblies and subassemblies.

a. The contractor's quality program or detailed inspection plan shall provide assurance of compliance of all characteristics with the applicable drawings and specifications utilizing as a minimum the conformance criteria specified in this document and the Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of QAP's, the applicable quality assurance provisions of MIL-F-13926 shall apply.

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b. Alternative inspection procedures, methods or equipment, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the procuring contracting officer for evaluation and approval by the Government. When required, the contractor shall demonstrate that the effectiveness of the proposed alternative(s) is equal to or better than the specified quality assurance provisions herein. In cases of dispute as to whether the contractor's proposed alternative(s) provide equal assurance, the provisions of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program plan or detailed inspection system, as applicable.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examinations and tests related to section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with the sampling plans specified in 4.4.2.3 herein. Examination and test for packaging and marking shall be in accordance with Section 5 herein. The tabulated classification of defects in 4.4.2.3 shall constitute the minimum inspection to be performed by the contractor after first article approval and prior to Government acceptance or rejection by item or lot.

4.4.2.3 Functional tests. The tests in Table X shall be performed on a 100% basis.

TABLE X. Functional tests (100 percent inspection).

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
101. Electrical	3.4	4.7.2
102. Performance	3.5	4.7.3
Minor:	None	N/A

4.5 Special sampling. One computer out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests of Table X and shall then meet the tests in Table XI. Unless otherwise specified in the contract, the items shall be returned to the lot upon successful completion of testing.

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TABLE XI. Special sampling.

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
103. Vibration	3.3.1	4.7.1.1
104. Shock	3.3.2	4.7.1.2
105. Altitude	3.3.3	4.7.1.3
106. Temperature	3.3.4	4.7.1.4
107. Humidity	3.3.5	4.7.1.5
Minor:	None	N/A

4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract.

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit

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and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2, the test equipment listed in Table XII (Test equipment list) shall be used in the performance of the applicable test methods and procedures as specified in 4.7.

TABLE XII. Test equipment list.

ITEM NUMBER		DESCRIPTION
1.	Power supply	6.5 ± 0.5 Vdc, regulated; ac ripple less than 50 mV peak-to-peak; capacity 3 A.
2.	Power supply	Dual, 33.0 ± 0.25 Vdc supplies, regulated; ac ripple less than 50 mV peak-to-peak; capacity 1 A.
3.	Power supply	24.0 ± 0.5 Vdc, regulated; ac ripple less than 100 mV peak-to-peak; capacity 100 mA.
4.	60 MHz oscillator	60.0 ± 5.0 MHz sine wave, 2.5 to 4.0 V, P-P output.
5.	Variable phase generator	Provide two output sine wave signals, one as reference phase and one with 0 to 360 degrees adjustable phase lag with respect to the reference phase. Frequency 1.0 ± 0.1 KHz. Power output 40 mW (5 V across 600 ohms). DC offset shall be adjustable to -0.5 ± 0.1 Volts.
6.	Dc digital voltmeter	Visual display 5 digits. Full scale range: 50,000 V or greater. Input resistance: 10 megohm or greater. Accuracy: ±0.01 percent.
7.	Oscilloscope	Dual channel; rise time of scope plus measuring probe, 10 ns/cm or less; input resistance 8 megohms or greater.
8.	Test fixture	See Figure 3. See Table XIII for test fixture parts list.
9.	Remote BIT tester assembly	TS-3211/TSM-100B, NSN 4931-01-212-2436, Drawing 9361217, or equivalent.
10.	Cable assembly	NSN 4933-01-210-9302, Drawing 9361225, or equivalent.

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TABLE XIII. Test fixture parts list.

DESIGNATION	NOMENCLATURE AND DESCRIPTION
DS1 thru DS4	Lamp, miniature bayonet, type 47
DS5	Lamp, miniature bayonet, type 313
E1, E2, E7 thru E12	Binding post, H. H. Smith 257 or equivalent
F1	Fuse, type 3AG, 3 A
F5, F6	Fuse, type 3AG, 1A
J1 thru J6	Connector, receptacle, UG-625B/U or equivalent
P1	Connector, plug, Burndy L18 TF32S6N-A
P4	Connector, plug, Bendix PTSF06SE22-78C
P5	Connector, plug, Burndy L12 TF106PX-A
R1	Resistor, 150 ohm, RL07S151G or equivalent.
S1	Switch, 4PST
S2	Switch, SPST
TP1 thru TP4	Tip jack, H. H. Smith - 1501 or equivalent
XDS1 thru XDS5	Lamp holder, Dialight 502-8136-102
XF1, XF4 thru XF6	Fuse holder, Littelfuse 342014

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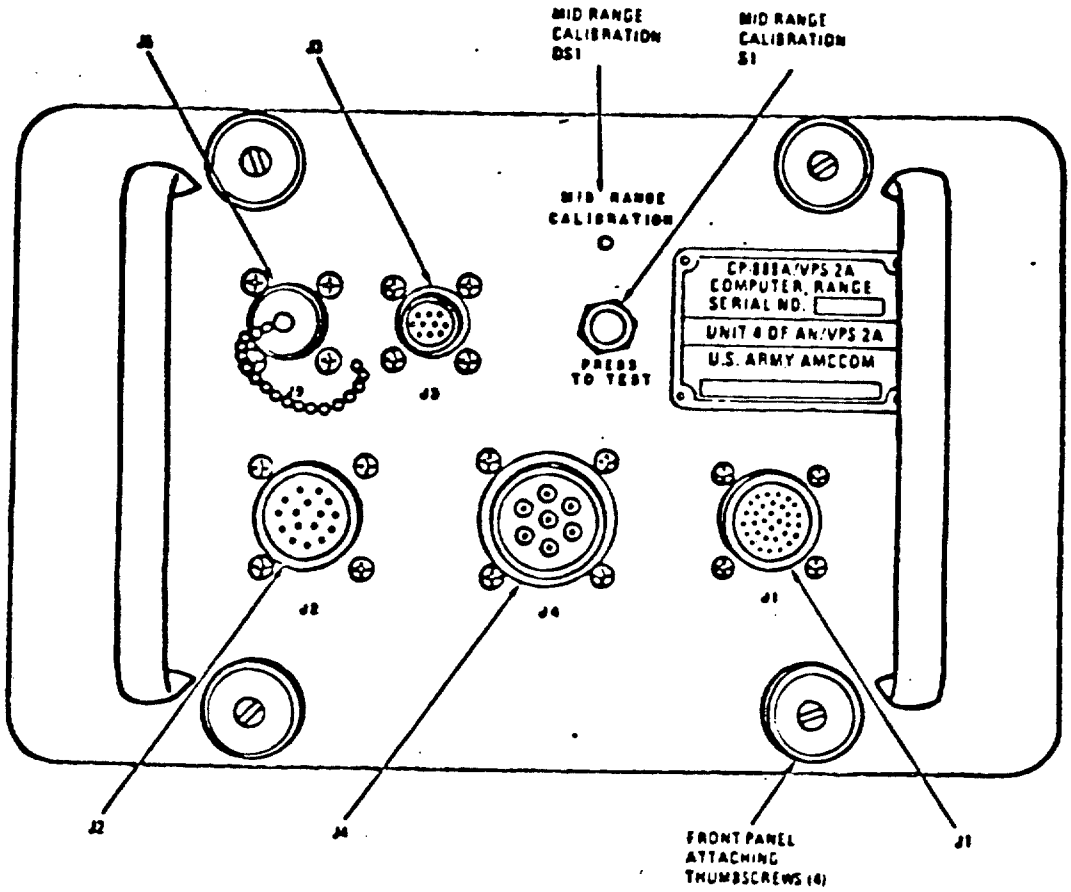


FIGURE 1. Range Computer.

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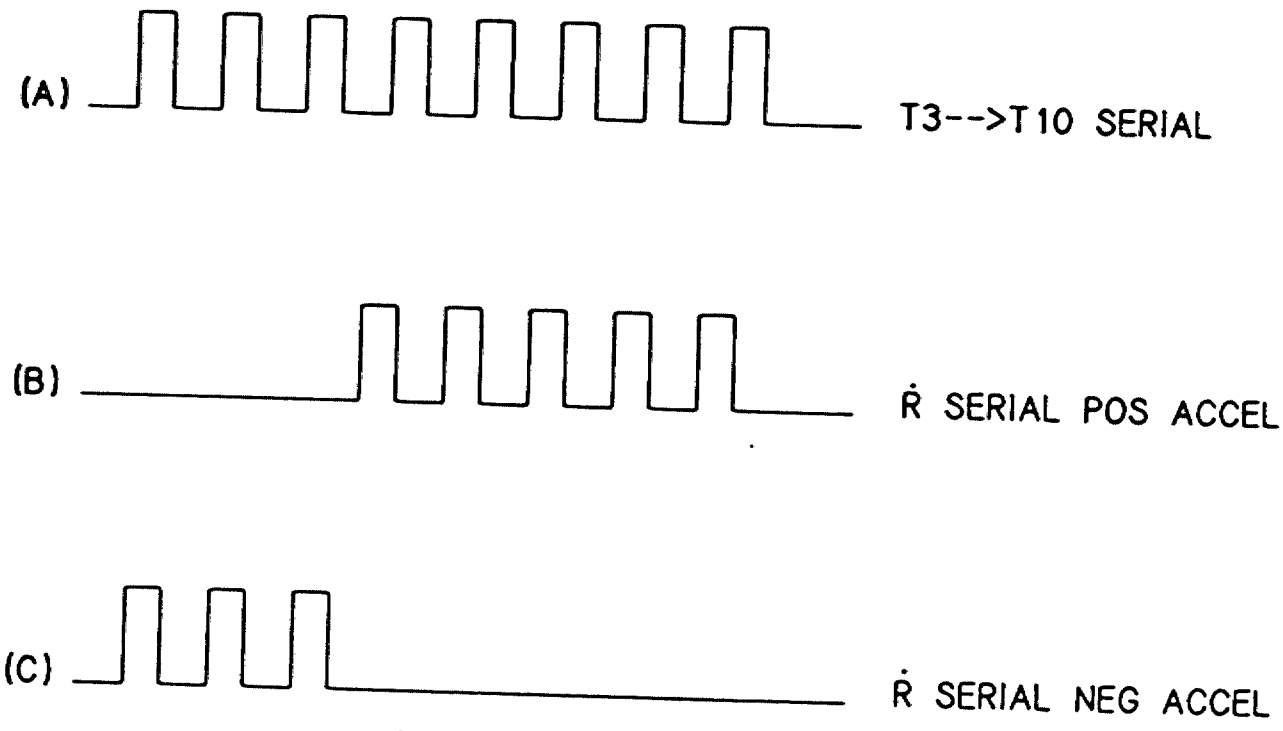


FIGURE 2. T3-> T10 and $\dot{R}$ serial signals.

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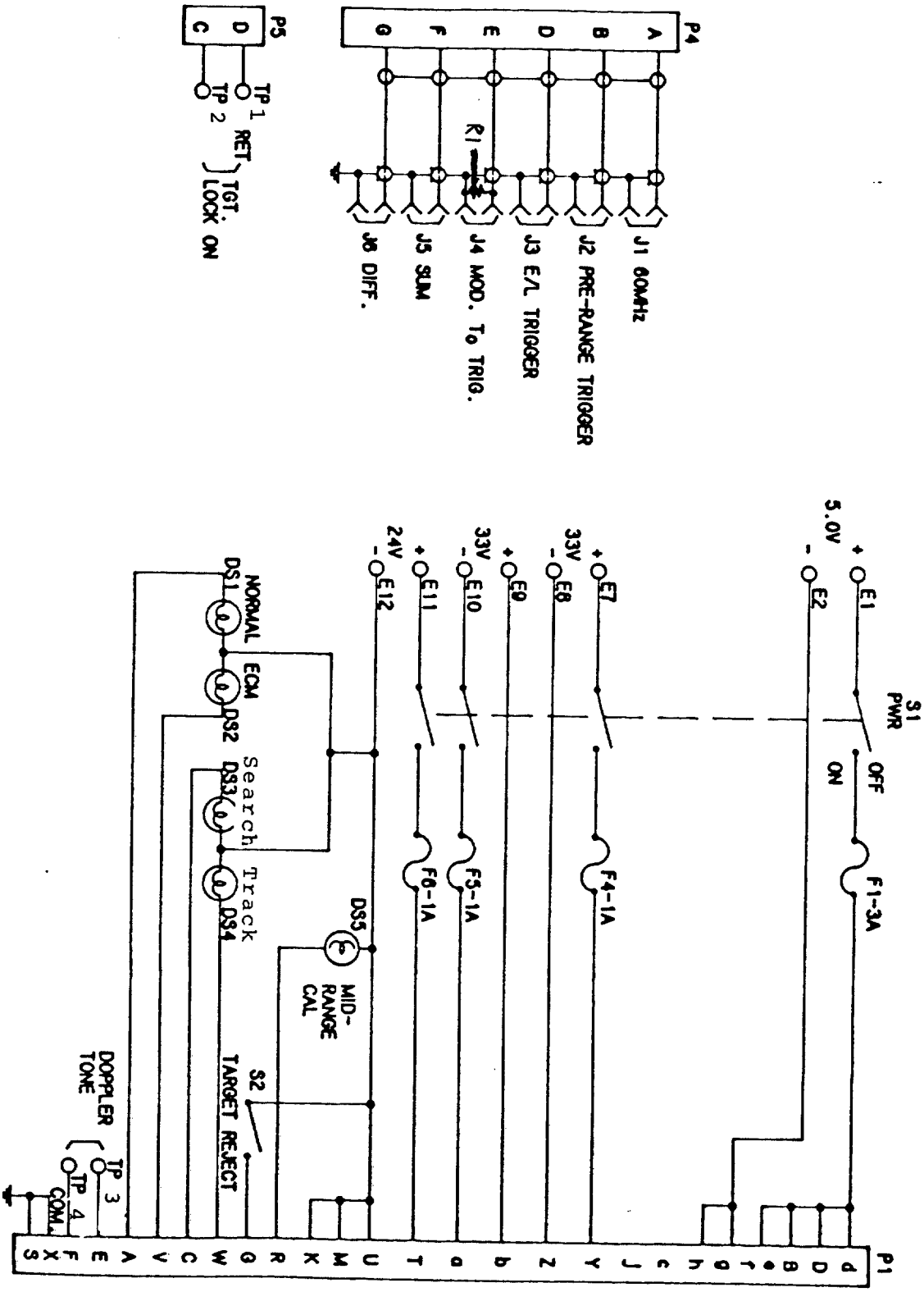


FIGURE 3. Test fixture.

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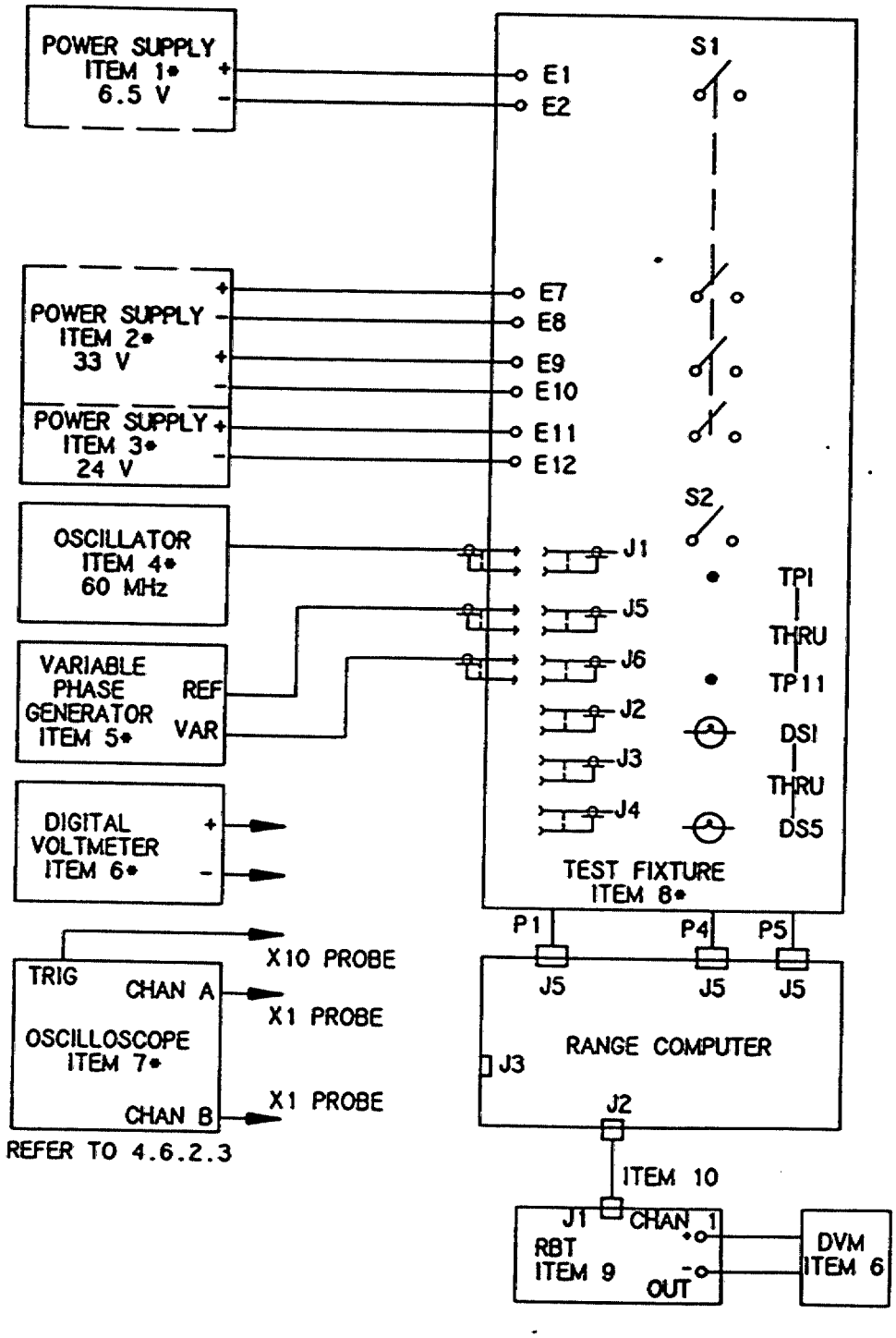


Figure 4. Test Setup

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4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance, and individual test methods of MIL-STD-810, except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

4.7.1.1 Vibration. The test shall be performed as specified in Method 514, Category 8, Procedure I, of MIL-STD-810 and paragraph 3.3.1 herein. In Step 1 of Procedure I only steps 4 through 8 of paragraph II-2.1 apply. The computer shall be attached to the vibration generator directly or with a fixture, and securely held by its normal means of attachment. After vibration, the computer shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1.

4.7.1.2 Shock. This test shall be performed as specified in Procedure I, Section II of Method 516 of MIL-STD-810 except that:

a. The post test functional check (Step 5 of Procedure I) need not be conducted until completion of all shock impulses.

b. The shock pulses shall be 15 g's ± 10% amplitude and 11 ms ± 10% duration.

c. The maximum g shall occur at 5.5 milliseconds ± 10%.

d. The shock measuring system shall have a flat frequency response within ± 10% from 20 to 2,000 hertz.

After the application of all the shock impulses, the computer shall be tested as specified in 4.7.2 and 4.7.3 to determine compliance with 3.3.2.

4.7.1.3 Altitude operating. This test shall be performed as specified in Procedures II, Method 500 of MIL-STD-810. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3.

4.7.1.4 Temperature.

4.7.1.4.1 High (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 501 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

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4.7.1.4.2 Low (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 502 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.5 Humidity (non-operating). This test shall be performed in accordance with Procedure III, Method 507 of MIL-STD-810 except that in Step 7, checkout shall be performed at the conclusion of the test only, with the item at standard ambient conditions. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.5.

4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment and visual examination for compliance with 3.4.

4.7.3 Performance. The selected computers shall be tested as follows: Connect power supplies to the test fixture (Table XII, item 8) in accordance with Figure 4. Adjust power supplies (Table XII, items 1 through 3) to deliver input voltages as specified in Table I. Leave the 60 MHz oscillator (Table XII, item 4) and the variable phase generator (Table XII, item 5) disconnected from the test fixture (Table XII, item 8) at this time. For 4.7.3.1 through 4.7.3.24 the requirements of 3.5 shall apply. Oscilloscope (Table XII, item 7) will be used to observe all signals unless otherwise noted. When it is specified to observe signals at test points, it shall mean the test points on the computer unless otherwise noted.

4.7.3.1 Voltage inputs and distribution. Turn the test fixture PWR switch S1 to the ON position and using the digital voltmeter (Table XII, item 6) observe the distributed voltages at terminals as specified in Table II.

4.7.3.2 Midrange calibration lamp 4DS1. With the conditions of 4.7.3, observe that the midrange calibration lamp 4DS1 shall be dimly lit.

4.7.3.3 Midrange calibration switch 4S1. Using midrange calibration lamp DS5 on the test fixture, depress the midrange calibration switch 4S1 on the computer and check for compliance with 3.5.3.

4.7.3.4 7.5 MHz clock output. Connect the 60 MHz oscillator (Table XII, item 4) to J1 on the test fixture with the amplitude requirements of 3.5.4. At TP38 observe the 7.5 MHz clock signal as specified in 3.5.4.

4.7.3.5 $\overline{\text{PRF RESET}}$ signal. Synchronize scope from TP20 and at TP20 observe the $\overline{\text{PRF RESET}}$ signal as specified in 3.5.5.

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4.7.3.6 $\overline{\text{PRF}}$ signal. Synchronize scope from TP20 and observe at TP19 the $\overline{\text{PRF}}$ signal as specified in 3.5.6.

4.7.3.7 $\overline{\text{PRF CAL}}$ signal. Synchronize scope from TP20 and observe at TP19 the $\overline{\text{PRF CAL}}$ signal as specified in 3.5.7 with the midrange calibration switch 4S1 depressed.

4.7.3.8 $\overline{\text{PRF}}$ signal. Synchronize scope from TP20 and observe at TP31 the $\overline{\text{PRF}}$ signal as specified in 3.5.8 with the midrange calibration switch 4S1 depressed.

4.7.3.9 $\overline{\text{T1}}$ signal. Synchronize scope from TP20 and observe at TP21 the $\overline{\text{T1}}$ signal as specified in 3.5.9.

4.7.3.10 Mod T_o TRIGGER signal. Synchronize scope from TP31 and observe at TP45 and also at J4 on the test fixture the MOD T_o TRIGGER signal as specified in 3.5.10.

4.7.3.11 T2 signal. Synchronize scope from TP21 and observe at TP34 the T2 signal as specified in 3.5.11.

4.7.3.12 T3 signal. Synchronize scope from TP21 and observe at TP23 the T3 signal as specified in 3.5.12.

4.7.3.13 $\overline{\text{T11}}$ signal. Synchronize scope from TP20 and observe at TP32 the $\overline{\text{T11}}$ signal as specified in 3.5.13.

4.7.3.14 $\overline{\text{T1/250}}$ signal. Synchronize scope from TP9 and observe at TP9 the $\overline{\text{T1/250}}$ signal as specified in 3.5.14.

4.7.3.15 $\overline{\text{T3/4KHz}}$ signal. Synchroize scope from TP10 and observe at TP10 the $\overline{\text{T3/4KHz}}$ signal as specified in 3.5.15.

4.7.3.16 $\overline{\text{200M}}$ signal. Establish the conditions of 3.5.16. Using switch S2 on the test fixture to apply the +24 V to 471-G, observe the $\overline{\text{200M}}$ signal at TP11 as specified in 3.5.16.

4.7.3.17 T3 --> T10 serial and $\dot{\text{R}}$ serial signals. Sequentially place switches 4S2 through 4S6 as specified in Table III. At TP17 and TP18 observe the T3 --> T10 serial and $\dot{\text{R}}$ serial signals, respectively, as specified in 3.5.17.

4.7.3.18 TRACK signals.

4.7.3.18.1 No target condition. As specified in 3.5.18.1 observe the TRACK signal at TP15 and TP24.

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4.7.3.18.2 Target condition. As specified in 3.5.18.2 observe the TRACK signal at TP15 and TP24.

4.7.3.18.3 Target fade (short term). As specified in 3.5.18.3 observe the TRACK signal at TP15 and TP24.

4.7.3.18.4 Target fade (long term). As specified in 3.5.18.4 observe the TRACK signal at TP15 and TP24.

4.7.3.18.5 Target lock-on signal. As specified in 3.5.18.5 observe the target lock-on signal at TP2 (signal) and TP1 (signal return) on the test fixture.

4.7.3.19 MIDRANGE lamp 4DS1. Observe the conditions of the midrange lamp as specified in 3.5.19.

4.7.3.20 E/L COARSE signal. Observe at TP33 the E/L COARSE signal as specified in 3.5.20.

4.7.3.21 PRE RANGE TRIGGER signal. Observe at TP36 and also J2 on the test fixture the PRE RANGE TRIGGER signal as specified in 3.5.21.

4.7.3.22 E/L TRIGGER.

4.7.3.22.1 E/L TRIGGER characteristics. Observe at TP43 and TP44 and also J3 on the test fixture the E/L TRIGGER characteristics as specified in 3.5.22.1.

4.7.3.22.2 E/L TRIGGER movements. Sequentially position switches 4S2 through 4S6 as specified in Table V and observe the E/L TRIGGER characteristics at TP43 and TP44 as specified in 3.5.22.2.

4.7.3.22.3 E/L TRIGGER leading edges. With the conditions of 3.5.22.3 observe the E/L TRIGGER leading edges at TP43, TP44 and J3 on the test fixture as specified in Table VI.

4.7.3.23 R> $\pm$ 10 M/S signal. Position switches 4S2 through 4S6 as specified in 3.5.23. Sequentially perform the steps of Table VII and observe the R> $\pm$ 10 M/S signal at TP42 as specified in Table VII.

4.7.3.24 Doppler signals response. Connect the variable phase generator (Table XII, item 5) as shown in Figure 3. Sequentially apply the sum and difference signals as specified in Table VIII under the conditions as specified in 3.5.24. Observe that the E/L TRIGGER at TP44 responds as specified in Table VIII.

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4.7.3.25 Subclutter visibility and range accuracy tests. After performing the tests of 4.7.3 through 4.7.3.24, substitute the computer into a Radar Set AN/VPS-2A and perform the subclutter visibility and range rate accuracy tests as specified in MIL-R-70560. The Radar Set AN/VPS-2A and test facilities used shall be as specified in MIL-R-70560.

4.7.3.26 Remote BIT assembly. The remote BIT assembly shall have met the requirements of MIL-C-70571 and MIL-C-70572, respectively. They shall be tested as follows:

a. Set the remote BIT tester switches as follows:
   (1) Power ON.
   (2) Send UP.
   (3) Normal/U4 test NORMAL.
b. Set station 1234 thumbwheel switches per Table XIV (Computer unit 4, BIT circuit card 4A16/4A17, voltage and signal output). Channel 1 and Channel 2 outputs shall be per Table XIV for each of the switch settings.

4.7.3.27 Remote bit voltage and signal output. With the conditions of para 3.6.3 in effect, a 60 MHZ signal applied as specified in para. 3.5.4, and with the unit 4 switches set - "Mode to Norm", "Accel to Pos 1", "Return to NO TGT", and "Range to Run", - Set the station codes and observe the results for steps 1 through 16 of Table XIV using the remote bit tester, PN 9361217.

4.7.3.28 Switch Controls. With the conditions of para 3.6.4 in effect, a 60 MHZ signal applied as specified in para 3.5.4, and with the unit 4 switches set - "Mode to Norm", "Accel to Neg", 1 "Return to TGT" and "Range to Run:, - Set the station codes and observe the results for steps 1 through 17 of Table XV using remote bit tester, PN9361217.

4.8 Soldering. Soldering processes shall be inspected by certified inspectors to the requirements and acceptance criteria of Note 2 of Drawing 12950877 as specified in 3.7.

5. PACKAGING

5.1 General. Preservation, packing and marking shall be in accordance with Special Packaging Instructions PS9361331.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The range computer CP-888A/VPS-2A is part of Radar Set AN/VPS-2A used in the PIVADS self-propelled and towed vehicles. The computer generates range and test data and transmits this information to the Fire Control Processor.

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TABLE XIV. Computer Unit 4, Bit Circuit Card 4A16/4A17, voltage and signal output.

STATION 1234 CODE	ANALOG CHANNEL	VOLTAGE OUT CH 1	COUNTER CHANNEL	CHANNEL 2 1 = ILLUMINATED 0 = OUT
1. 13 0 4 0	+15 Vdc Bit Supply	8.75 ± 0.25 Vdc	CLK 4 (EAT)	10100000
2. 13 1 4 1	+33 Vdc	9.69 ± 0.25 Vdc	_ R	See Note 1
3. 13 2 4 2	-33 Vdc	0.31 ± 0.05 Vdc	E/L TRIG OUT	01100110
4. 13 3 4 3	+25 Vdc	8.55 ± 0.25 Vdc	Re Scaled	01100110
5. 13 4 4 4	-25 Vdc	1.45 ± 0.20 Vdc	GND (Spare)	00000000
6. 13 5 4 5	+15 Vdc	8.75 ± 0.25 Vdc	GND (Spare)	00000000
7. 13 6 4 6	-15 Vdc	1.25 ± 0.10 Vdc	T3 - T10 (8)	01100110
8. 13 7 4 7	+6.5 Vdc	7.84 ± 0.20 Vdc	To TRIG	01100110
9. 13 8 4 8	+5 Vdc	7.50 ± 0.20 Vdc	$\overline{PRF}$	01100110
10. 13 9 4 9	MH Range Cal Relay	8.40 ± 0.5 Vdc	$\overline{T2}$	01100110
11. 13 10 4 10	+28 Vdc	8.40 ± 0.5 Vdc	$\overline{T11}$	01100110
12. 13 11 4 11	SUM SIGNAL	5.00 ± 0.05 Vdc	$\overline{T1}$	01100110
13. 13 12 4 12	GND (Spare)	5.00 ± 0.05 Vdc	E/L TRIG	01100110
14. 13 13 4 13	+5 Vdc Bit Supply	9.960 ± 0.025Vdc	E/L COARSE	01100110
15. 13 14 4 14	GND	5.00 ± .05 Vdc	CLK 5 (EAT)	01100110
16. 13 15 4 15	-15 Vdc Bit Supply	1.25 ± .05 Vdc	CLK 5 (EAT)	01010000

NOTE 1: Place Unit 4 MODE Switch (4S2) to TEST and RETURN switch (4S4) to TGT. Channel 2 will display 00000000. Place Unit 4 RETURN switch (4S4) to NO TGT. Channel 2 should display 00000000 for approximately 3 seconds and then display 00010100. Place Unit 4 MODE switch (4S4) to NORM.

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TABLE XV. Computer Unit 4, BIT Circuit Card 4A16/4A17, Switch Control Checks.

	STATION 1234 CODE	CHANNEL 1 READING 1 - 8	CHANNEL 2 READING 9 - 16	
				(See Note 1)
1.	14 0 14 0	11001021	01110010	
2.	15 0 15 0	11001221	01100011	
3.	15 1 15 1	11201221	01100001	
4.	15 2 15 2	11201221	01100001	
5.	15 3 15 3	11201221	01110001	
6.	15 4 15 4	11001111	01101011	
7.	15 5 15 5	11001221	01100001	
8.	15 6 15 6	10211111	01111011	(Note 2)
9.	15 7 15 7	11201221	01110001	
10.	15 8 15 8	11201221	01100111	
11.	15 9 15 9	11201221	01100001	
12.	15 10 15 10	11201221	01110111	
13.	15 11 15 11	11201221	01110001	
14.	15 12 15 12	11201111	01101111	
15.	15 13 15 13	11201221	01100001	
16.	15 14 15 14	10211111	01111111	(Note 2)
17.	15 15 15 15	11001221	01110001	

NOTE 1. Apply .2 Vpp to J4-G and less than .1Vpp to J4-F for Steps 1 through 17.

NOTE 2. Lights 6 and 7 in Channel 1 will blink for 40 milliseconds every 17 seconds.

FOOTNOTES: Indicator OFF = 0, Indicator LIT = 1, Indicator Blinking = 2

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).
c. Applicable stock number.
d. First article sample requirements.
e. Packaging requirements, if other than specified in Section 5.
f. Serialization requirements, if applicable.
g. Certification of conformance for each lot of shipment of product.

6.3 Submission of contractor inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army AMCCOM, Attn: SMCAR-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.5 Subject term (Key word) listing.

Fire Control System
Product Improved Vulcan Air Defense System (PIVADS)
Remote Bit Assembly
Self-propelled M163A2
Towed M167A2

6.6 Changes from previous issue. The previous issues were for ARDEC internal use only. They were not forwarded to Naval Publication and Forms Center (NPFC) for publication. This issue is the initial public release for unlimited distribution.

Custodian:
Army - AR

Preparing activity:
Army-AR

(Project 1285-A743)

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4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

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c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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